

Time Series Analysis and Forecasting of Monthly Road Traffic Accidents

Independent Research Project

Author: Anastasia Demidova

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Abstract

This study investigates the ability of classical time series models to capture and forecast the dynamics of road traffic accident data. Using seasonal decomposition and a SARIMA model, the analysis identifies seasonal patterns and strong autoregressive dynamics. The results are further evaluated against a baseline model, highlighting both the strengths and limitations of classical approaches and motivating the need for more flexible modelling frameworks.

1. Introduction

Road traffic accidents remain a major public safety concern worldwide, with significant social and economic consequences. Understanding temporal patterns in accident data is essential for effective policy planning, risk assessment, and the design of targeted safety interventions.

Time series analysis provides a statistical framework for modelling and forecasting accident counts over time. Identifying seasonal patterns and persistence in accident data may contribute to improved allocation of resources and more informed decision-making in traffic safety management.

The aim of this study is to analyse the temporal dynamics of monthly road accident counts and to develop a statistical model for short-term forecasting. In particular, the analysis focuses on identifying trends, seasonal patterns, and autoregressive structures in the data using classical time series methods.

The central research question of this study is whether classical univariate time series models are sufficient to capture the dynamics of road traffic accident data, or whether more flexible approaches incorporating external factors are required for accurate modelling and forecasting.

2. Data

The dataset contains monthly counts of road traffic accidents from January 2020 to December 2024. The data therefore consist of 60 observations representing a five-year period.

Monthly aggregation allows the identification of seasonal patterns and longer-term trends in accident counts.

The dataset was obtained from publicly available road safety statistics published by the UK Department for Transport. The analysis was implemented in Python using the pandas, matplotlib, and statsmodels libraries.

3. Exploratory Analysis

An initial visualisation of the time series reveals substantial variation in accident counts over time. A pronounced drop in accident numbers is observed in early 2020, likely reflecting reduced mobility during the COVID-19 pandemic.

After this period, accident counts gradually recover and fluctuate around a relatively stable level between approximately 7500 and 9500 accidents per month.

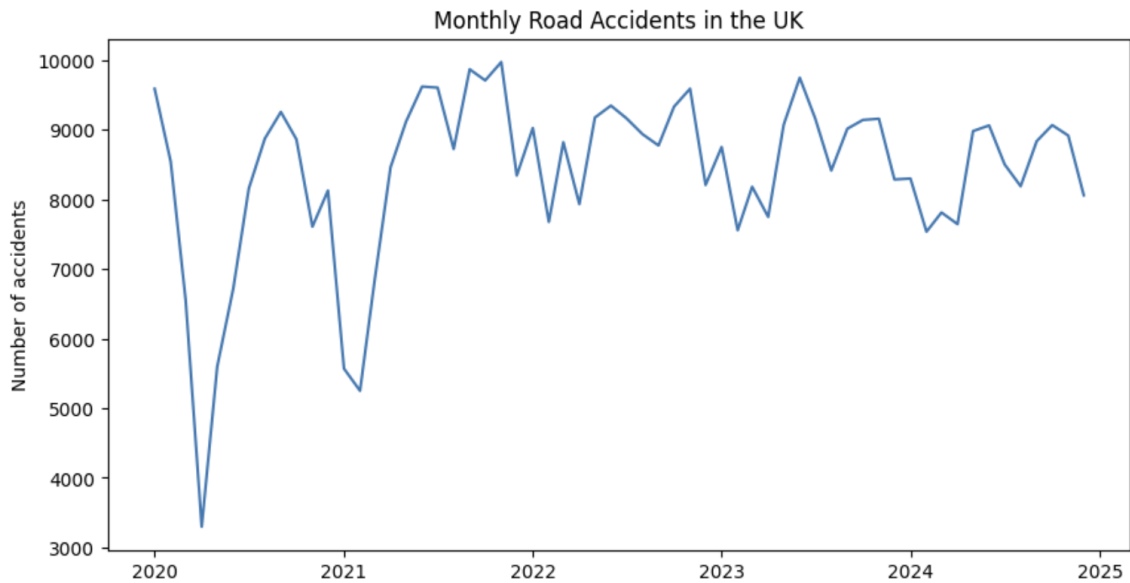


Figure 1. Monthly number of road accidents (2020–2024)

4. Time Series Decomposition

To better understand the structure of the time series, seasonal decomposition was applied. This method separates the observed series into three components: trend, seasonal variation and residual noise.

The decomposition reveals a clear seasonal pattern. Accident counts tend to increase during late spring and summer months and decrease during winter. The trend component indicates a temporary decline in 2020 followed by a recovery in subsequent years.

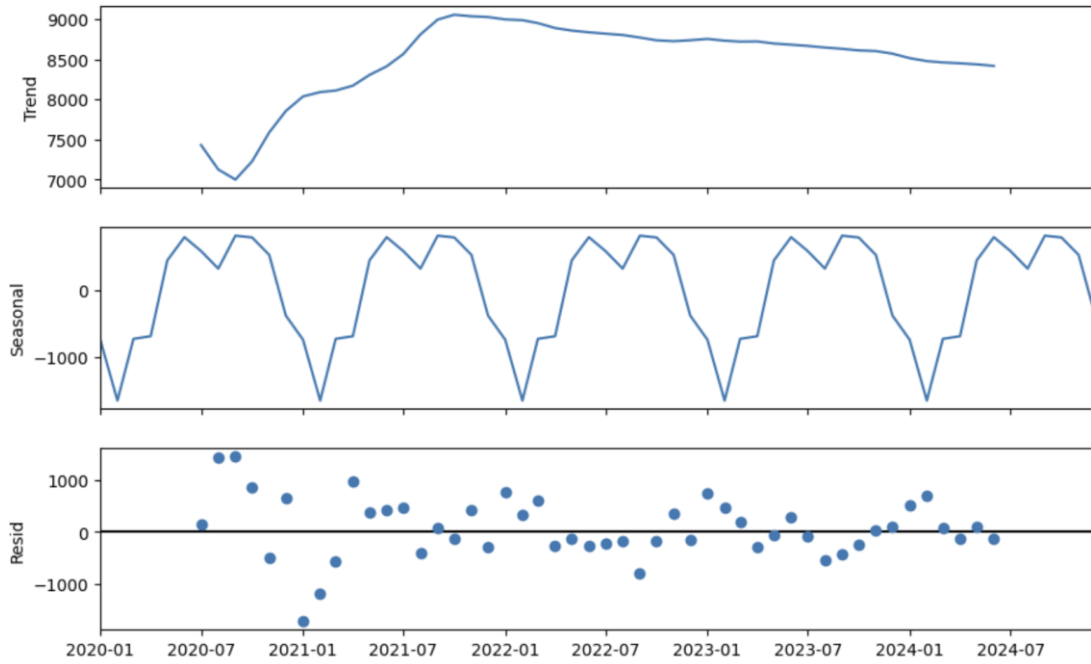


Figure 2. Seasonal decomposition of the time series

5. Stationarity Test

Stationarity of the time series was evaluated using the Augmented Dickey–Fuller test. The test statistic was -4.79 with a p-value below 0.001 . This result allows rejection of the null hypothesis of a unit root, suggesting that the series is stationary.

6. Model Identification

Autocorrelation (ACF) and partial autocorrelation (PACF) plots were used to identify the autoregressive structure of the series. The ACF shows gradual decay, while the PACF displays a strong spike at lag 1, indicating a dominant autoregressive component.

Based on these diagnostics, a seasonal autoregressive integrated moving average model was selected.

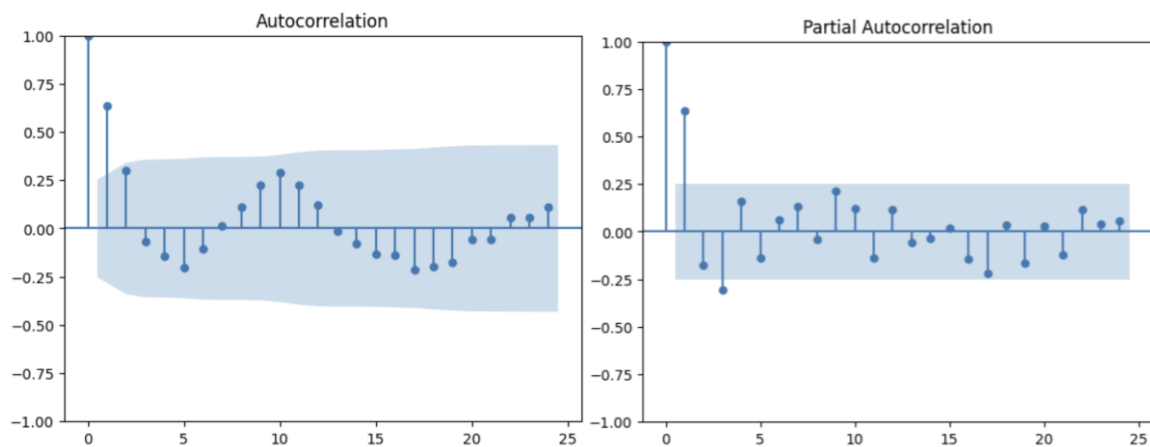


Figure 3. Autocorrelation and partial autocorrelation functions

7. Model Estimation

A SARIMA(1,0,0)(1,0,0,12) model was estimated to capture both short-term dependence and seasonal structure in the data. The autoregressive coefficient at lag 1 is highly significant ($p < 0.001$), indicating strong persistence in monthly accident counts.

Diagnostic tests suggest that residual autocorrelation is not statistically significant according to the Ljung–Box test. The Jarque–Bera test does not reject the assumption of normally distributed residuals.

Model selection was guided by ACF/PACF diagnostics and standard statistical criteria.

8. Forecasting

The estimated model was used to forecast accident counts for the following twelve months. The forecast suggests that accident counts are expected to remain relatively stable, with predicted values around 7500–8000 accidents per month.

The confidence interval widens for longer forecast horizons, reflecting increasing uncertainty in the predictions.

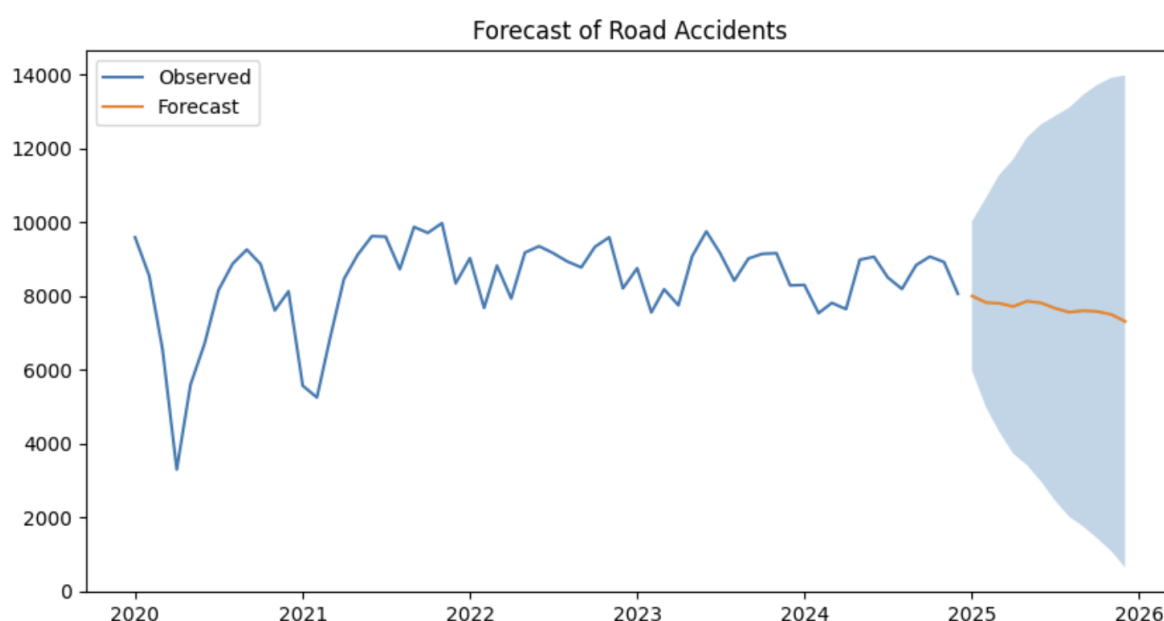


Figure 4. Forecast of monthly accident counts for the next 12 months

9. Model Comparison

To assess the performance of the SARIMA model, its forecasts were compared with a simple baseline model based on a naive forecasting approach, where future values are assumed to be equal to the most recent observation.

The SARIMA model demonstrates improved performance in capturing seasonal patterns and short-term dynamics in the data. In particular, it provides smoother forecasts and better reflects recurring seasonal fluctuations compared to the baseline model. However, the difference in predictive performance decreases for longer forecast horizons, indicating increasing uncertainty in both approaches.

These results suggest that while classical time series models offer improvements over simple baselines, their predictive capabilities remain limited, particularly in the absence of additional explanatory variables.

Forecast accuracy was qualitatively assessed, with potential for further evaluation using metrics such as MAE or RMSE.

10. Limitations

This study has several limitations. First, the analysis is based on a univariate time series model and does not incorporate potentially relevant explanatory variables such as traffic volume, weather conditions, or policy interventions. As a result, the model may fail to capture important drivers of accident dynamics.

Second, the dataset covers a relatively short time period, which may limit the robustness of long-term forecasts and the stability of estimated parameters.

Finally, classical time series models assume linear relationships and may not adequately capture more complex or nonlinear interactions present in real-world data.

11. Conclusion

This study demonstrates how classical time series methods can be applied to analyse and forecast road traffic accident data. The analysis identifies clear seasonal patterns and strong autoregressive dependence in monthly accident counts, and the SARIMA model provides reasonable short-term forecasts.

However, the results also highlight important limitations of univariate time series models, particularly their inability to incorporate external factors that may influence accident dynamics. These limitations suggest that more flexible modelling approaches may be required for improved predictive performance.

Future research could extend this analysis by incorporating exogenous variables such as weather conditions, traffic flow, or policy interventions into the modelling framework. In addition, machine learning approaches, including tree-based models or recurrent neural networks (e.g., LSTM), could be explored to capture nonlinear relationships and interactions between multiple influencing factors.

Such extensions may contribute to a more comprehensive understanding of traffic safety dynamics and support the development of more effective data-driven policy measures.

This highlights the importance of integrating both statistical rigour and data-driven approaches in modern traffic safety research.

12. References

- Department for Transport (UK). *Road Safety Data – Collisions dataset*. Available at: <https://www.gov.uk/government/statistics/road-safety-statistics>.
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